

Mathematics Curriculum Area

Statistical Investigation

Create an informative poster presentation based on the collection and interpretation of data towards addressing a practical research question.

Name _____ Due date _____

Year _____ Class _____ Teacher _____

The poster will be completed in a small group of 2-3 people and all posters will be entered into a national competition. You are to create a poster (up to 84cm by 59cm) on a topic of your own choice. The poster may be presented in either a hard copy or a digital copy. It should be readable from 2 metres away. Your posters will be presented at a parent evening early in Term 4 before being sent away for judging in the competition

Tasks

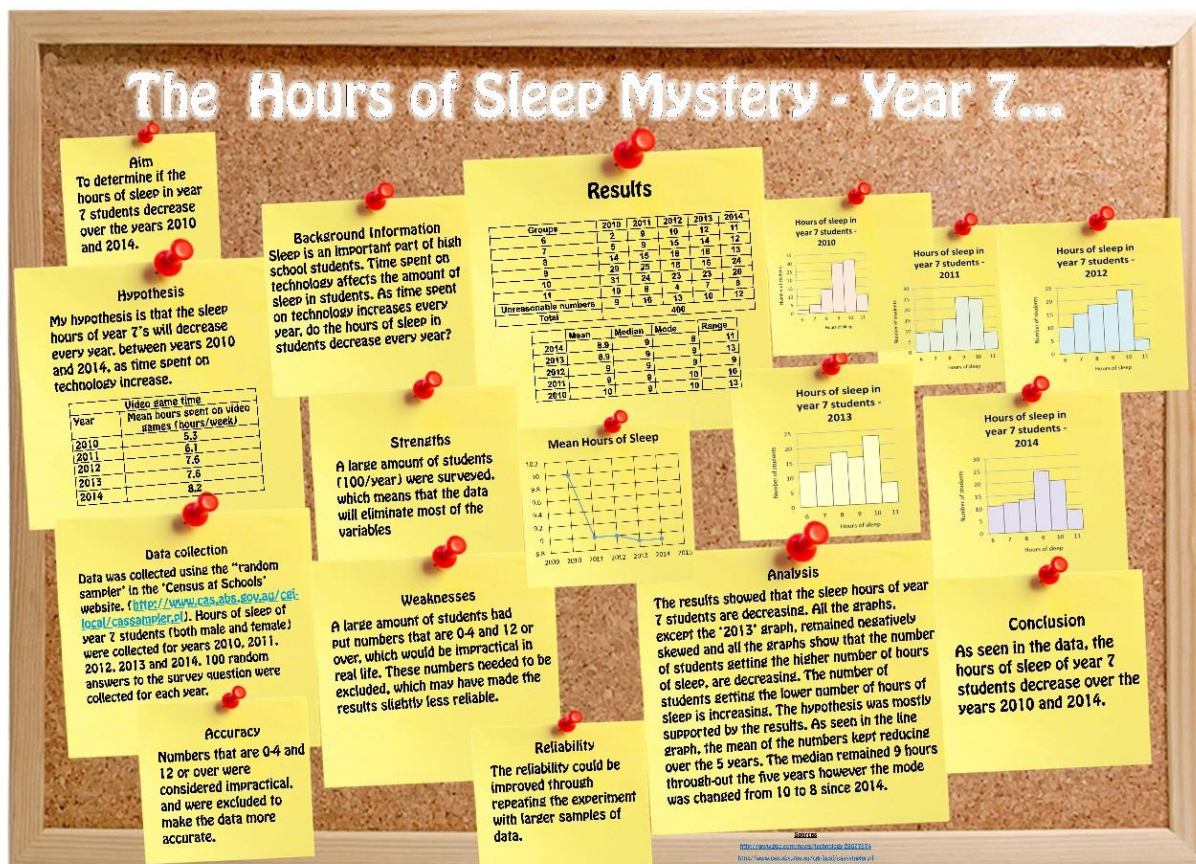
1. Choose a topic question that you will collect data on.
2. Complete a plan of action. Use the planning document attached to guide your project to completion.
3. State your aim, hypothesis and give some background information.
4. Collect the data you require. You may use either primary or secondary data. This may include completing your own experimentation or conducting a survey.
5. Produce the data in a table of results.
6. Display your data in appropriate graphs.
7. Analyse the data using appropriate techniques.
8. Discuss the findings of your data analysis clearly outlining any possible biases or outliers and how you have dealt with these issues.
9. Complete your conclusion.
10. References

Please refer to attached Marking Rubric for school assessment and for further information about competition rules see <https://www.ssaipostercomp.info/>

You are free to choose a topic from any subject (current events, psychology, music, art, forensic anthropology, bio-chemistry, nutrition, marketing, education, etc) as long as you can **collect data and use statistic methods to analyse the data you collect**. To help you get started, here are some topic suggestions.

- Does talking on your mobile phone affect concentration on surrounding activity?
- Does music affect concentration or mood?
- How does the nutritional value of breakfast cereals compare?
- Do soft drinks affect concentration?
- Can a person's height be predicted from body part measurements?
- Are bacteria levels different under toe nails and finger nails?
- Does location of toothbrush affect bacteria levels?
- Do people notice differences between Lift and Solo?
- Does a particular learning technique work best when studying?
- Does Vitamin C level differ between organic and non-organic orange juices?

Example



POSTER CHECKLIST

1. Clarity of the message:

- a. Is the poster understandable without extra information?
- b. Are the objectives, research question or hypotheses clear?
- c. Are the results and conclusions clear?
- d. Is the poster clearly and logically set out?

2. Data collection:

- a. Are the collected data appropriate for answering the research question?
- b. Are the data collection methods clearly stated? (e.g., primary data collection by print, telephone, web survey, or other measurement tools; or secondary sources are cited — published reports, databases etc.)
- c. Has quality of the data been considered? (i.e., accuracy of measurements, size of the sample, reliability of sources, methods of sampling, etc.)
- d. Has variation been considered when obtaining data?
- e. Have comparison groups been considered and well-constructed where applicable?
- f. Are the data sources referenced in the poster?

3. Graphs and tables:

- a. Are the graphs/tables/statistics appropriate for displaying and summarising the data?
- b. Are there adequate graphs/tables from different perspectives?
- c. Does every graph/ table add something to the poster?
- d. Are the graphs/tables properly titled and explained?

4. Calculations

- a. Have you included appropriate statistical calculations?

5. Analysis and conclusions

- a. Are the data analysed in terms of the research question or hypotheses?
- b. Is the analysis appropriate for the kind of data collected?
- c. How well is the research question answered?
- d. Are there conclusions and are they supported by the data?
- e. Are there any limitations discussed or improvements for future studies suggested?

6. Presentation:

- a. Is the poster readable from 2 metres (7 feet) away?
- b. Is there a good balance between graphs and text?
- c. Does the poster look neat?

7. Creativity/importance:

- a. Is the research question creative, original?
- b. Can the study answer an interesting question?
- c. Is the design creative and original?
- d. Is the poster eye-catching?